MoreDiff: High-Fidelity Motion Refinement via Manifold Rectification

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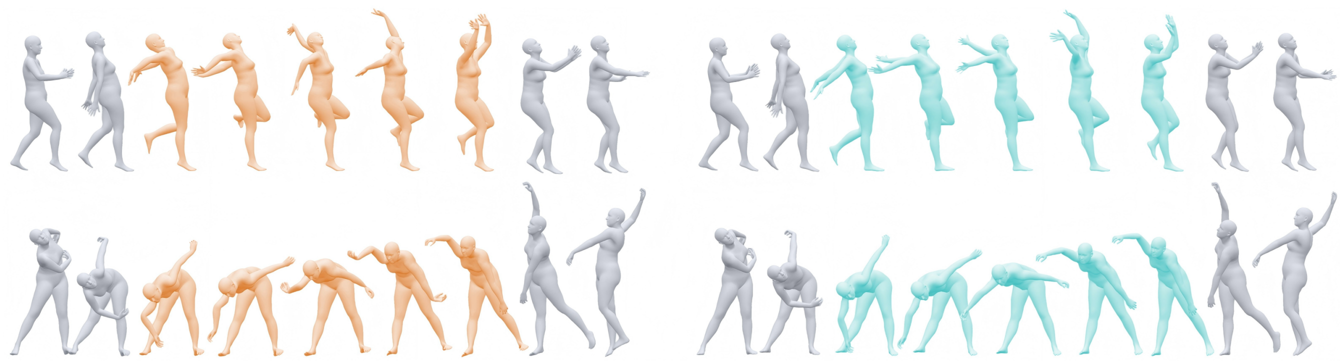

Fig. 1. Motion refinement by MoreDiff. Top: corrupted and refined results on monocular video-based mocap; Bottom: results on optical marker-based mocap. MoreDiff provides a novel manifold rectification-based approach for high-fidelity motion refinement. Unlike inpainting-style methods that require extremely large diffusion steps (e.g., > 700) and inevitably introduce fidelity degradation, MoreDiff only needs a small number of steps (e.g., ≤ 10), which fundamentally avoids fidelity loss and enables high-quality, high-fidelity motion refinement.

Motion capture data obtained in practical scenarios often contains corruptions such as joint pops, unnatural twists, tracking drift, foot penetration, and broken body configurations. Motion refinement aims to correct these errors while preserving the original performance, making it distinct from unconditional motion generation. Existing diffusion-based refinement methods usually formulate this task as motion inpainting, but they face a quality-fidelity dilemma: few denoising steps preserve the input but fail to remove severe corruptions, while many steps strengthen the motion prior but over-smooth the motion and cause fidelity loss.

We propose **MoreDiff**, a high-fidelity motion refinement framework that formulates motion cleanup as **manifold rectification**. Instead of relying on long denoising trajectories, MoreDiff learns a step-agnostic Manifold Rectification Gradient from q-sampled corrupted motions with clean-motion supervision, enabling few-step correction of structured off-manifold mocap errors. MoreDiff further combines corruption synthesis, geometric consistency regularization, and adaptive rectification to improve robustness and preserve reliable motion regions.

Experiments on synthetic, optical mocap, and monocular video mocap corruptions show that MoreDiff improves motion quality while better preserving the input compared with existing diffusion-based refinement methods. Although trained only with synthetic corruptions, MoreDiff generalizes well to real-world corruptions, offering an effective, efficient, and low-cost solution for practical motion capture post-processing and animation cleanup. **Our code and model weights will be released publicly**.

Additional Key Words and Phrases: Motion Refinement, Diffusion Model, Manifold Rectification

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1 Introduction

High-quality human motion data is essential for character animation, motion capture system development and humanoid robotics control. However, motion capture data obtained in practical scenarios is often imperfect. Optical systems may suffer from occlusions and marker swaps [Ghorbani and Black 2021], and monocular video-based estimators often produce jitter, depth ambiguity, and physically implausible poses [Shimada et al. 2020]. These errors appear as joint pops, unnatural twists, tracking drift, foot penetration, or broken body configurations, and usually require tedious manual cleanup by experienced artists.

Motion refinement aims to automate this cleanup process. Unlike unconditional motion generation, refinement has a stricter requirement: the output should be both plausible and faithful to the input. A desirable method should correct corrupted regions while preserving the timing, style, trajectory, and intent of the original performance. Therefore, motion refinement is not simply about generating a new plausible motion, but about accurately refineing an existing one with minimal unnecessary changes.

Recent diffusion-based methods have shown promise by formulating motion refinement as an inpainting problem. Given a corrupted sequence, they use the learned motion prior to resample unreliable regions and improve motion plausibility. However, this paradigm often suffers from a critical **quality-fidelity dilemma** (Fig. 2). With few denoising steps, the output remains close to the input but severe corruptions may remain. With more denoising steps, the model provides stronger refine, but often over-smooths the motion toward the learned manifold and overwrites input-specific details, resulting in

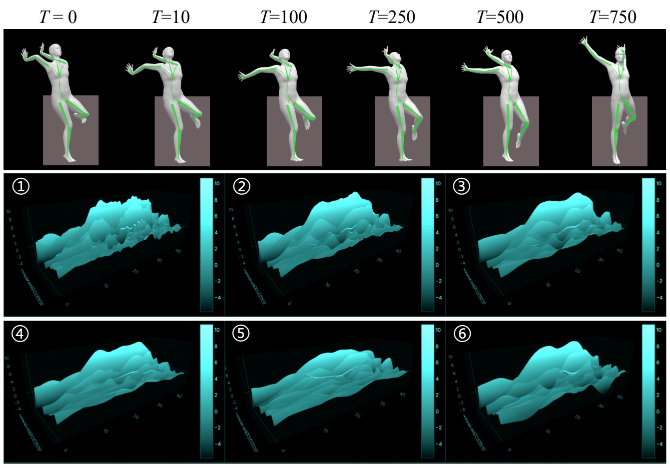

Fig. 2. Illustration of the quality-fidelity dilemma in diffusion-based motion refinement. More denoising steps improve refine, but the stronger motion prior tends to over-smooth the input toward the learned motion manifold, erasing motion details and reducing fidelity. Fewer steps better preserve the input, but often leave noticeable corruptions uncorrected.

fidelity loss. This behavior conflicts with the goal of motion cleanup, where reliable parts should be preserved.

We argue that this dilemma stems from the timestep-dependent rectification behavior of standard diffusion models. In diffusion-based refinement, the corrupted motion is first perturbed and then pulled back toward the clean motion manifold by the score-guided reverse process. However, the effective rectification strength of the standard score is strongly tied to the diffusion timestep. At small timesteps (e.g.,10), the input is only mildly perturbed, which preserves fidelity but provides weak correction for structured off-manifold corruptions. At large timesteps (e.g.,> 700),, the accumulated score guidance becomes strong enough to refine severe corruptions, but the input motion also loses more performance-specific information, causing the reverse process to rely heavily on the learned prior and overwrite details. Thus, standard diffusion refinement couples refine strength with denoising length, making it difficult to achieve both high quality and high fidelity.

To address this issue, we formulate motion refinement as **manifold rectification**. Clean human motions lie near a manifold of anatomically and physically plausible motion patterns, while corrupted mocap sequences deviate from this manifold in structured ways. Instead of relying on a long denoising trajectory to gradually impose the motion prior, our goal is to learn a direct rectification direction that pulls corrupted poses back toward their clean counterparts while preserving reliable motion regions.

Based on this view, we propose **MoreDiff**, a high-fidelity motion refinement framework based on step-agnostic **Manifold Rectification Gradient** (MRG). During training, we expose the diffusion model not only to q-sampled clean motions, but also to q-sampled corrupted motions, while supervising both with the corresponding clean motion. This teaches the model to produce non-trivial rectification signals across diffusion timesteps, allowing effective correction even with very few denoising steps. MoreDiff further combines corruption synthesis, geometric consistency regularization, and adaptive rectification to improve robustness and avoid unnecessary modifications to reliable regions.

We evaluate MoreDiff on synthetic corruptions, optical mocap corruptions, and monocular video-based mocap corruptions. Experiments show that MoreDiff improves motion quality while better preserving the input motion compared with existing diffusion-based refinement methods. Although trained only with synthetic corruptions, MoreDiff generalizes to real-world capture corruptions, providing an effective, efficient, and low-cost solution for motion post-processing in animation pipelines.

Overall, our contributions are as follows:

- We identify and analyze the quality-fidelity dilemma in diffusion-based motion refinement from a score-guided rectification perspective. We show that standard diffusion couples rectification strength with denoising length, and formulate motion refinement as manifold rectification to enable few-step correction of structured off-manifold corruptions without unnecessarily regenerating the entire motion.
- We propose MoreDiff, a high-fidelity motion refinement framework that learns step-agnostic Manifold Rectification Gradient from q-sampled synthesis corrupted motions with clean-motion supervision. Combined with geometric consistency regularization and adaptive rectification, MoreDiff enables targeted cleanup while preserving reliable motion regions.
- Experiment results show that training with well-designed synthetic corruptions is sufficient for common optical mocap and monocular video mocap corruptions. This makes MoreDiff an efficient and low-cost solution for practical motion capture post-processing and animation cleanup.

2 Related Work

2.1 Human Motion Priors and Diffusion Generation

Learning priors of natural human motion has long been a central problem in character animation. Recent diffusion-based motion generation methods have achieved impressive results in synthesizing plausible and diverse motions from noise or high-level conditions, such as text, trajectories, keyframes, or other control signals [Chen et al. 2023; Dai et al. 2024; Karunratanakul et al. 2023; Shafir et al. 2023; Tevet et al. 2022; Xie et al. 2023; Zhang et al. 2024b]. By modeling motion synthesis as a gradual denoising process, these methods provide powerful generative priors over human motion and have become a strong foundation for motion modeling.

However, motion refinement has a different objective from motion generation. Generation-oriented methods aim to synthesize a plausible motion sample, while refinement starts from an existing corrupted motion and should preserve its original timing, trajectory, style, and intent. Therefore, directly applying a strong generative prior to refinement may unnecessarily rewrite reliable parts of the input. This distinction is central to our work: we use diffusion not to regenerate the whole motion, but to provide targeted rectification for corrupted off-manifold regions.

2.2 Motion Capture Cleanup and Refinement

Motion capture data obtained in practical scenarios often contains corruptions caused by occlusions, marker swaps, sensor noise, tracking failures, or equipment limitations. Motion cleanup, also known as motion refinement, aims to correct these corruptions while preserving the captured performance [Holden 2018; Lin et al. 2023; Xiao et al. 2008]. Early methods mainly focus on marker-based mocap data, including marker trajectory denoising, marker labeling, and marker-to-skeleton reconstruction [Chen et al. 2021; Holden 2018; Pan et al. 2023, 2024]. These approaches are effective in marker-based production pipelines, but their assumptions on input markers and capture settings limit their applicability to skeleton-level motions produced by diverse systems, such as inertial mocap or monocular video-based estimators.

Another line of work improves motion plausibility through physical simulation or physics-aware control. By optimizing or imitating motions in a physics simulator, these methods can reduce physically implausible corruptions and improve dynamic consistency [Liu et al. 2010; Luo et al. 2023; Peng et al. 2018; Tessler et al. 2024; Truong et al. 2024; Yao et al. 2024; Yuan et al. 2023]. Nevertheless, physics-based correction relies heavily on accurate body models, contact handling, and control policies, and may still introduce jitter, foot sliding, or unstable behaviors when the simulated dynamics do not match the captured motion. Thus, physics alone is not always sufficient for high-fidelity cleanup in animation pipelines.

Data-driven kinematic methods directly learn to map corrupted motions to clean motions, which is closer to the refinement setting considered in this paper [Chen et al. 2021; Holden 2018; Li et al. 2024; Mall et al. 2017; Pan et al. 2023, 2024; Zhang et al. 2024a]. These methods can effectively correct many motion corruptions, but they often require paired corrupted-clean training data. Such pairs are expensive to obtain because clean targets usually require manual correction from experts. To avoid the labor intense clean targets obtaining, we using clean motion datasets and well-designed ad-hoc corruptions synthesis to create diverse clean-corrupted samples, and then focuses on whether a learned rectification behavior can generalize to real-world optical and video-based mocap corruptions.

2.3 Diffusion-based Motion Inpainting and Refinement

Recent motion inpainting methods use generative motion priors to fill missing, corrupted, or unreliable motion regions [Cohan et al. 2024; Guo et al. 2024; Hwang et al. 2025; Shafir et al. 2023]. Diffusion models are particularly attractive for this purpose because their iterative denoising process can impose a strong clean-motion prior and produce plausible completions. In the context of motion cleanup, diffusion-based methods have shown promising results by treating corrupted frames as regions to be resampled or refineed. StableMotion [Mu et al. 2025], for example, learns from mixed-quality motion data with quality indicators and demonstrates the potential of diffusion models for practical mocap post-processing.

Despite this progress, existing diffusion-based refinement methods usually inherit the sampling behavior of generation models. To remove severe corruptions, they often need a long denoising trajectory so that the generative prior becomes strong enough. However, a stronger prior may also over-smooth the input motion toward the learned motion manifold and overwrite reliable motion details, leading to fidelity loss. Conversely, using only a few denoising steps better preserves the input but often leaves structured off-manifold corruptions uncorrected. This quality-fidelity dilemma is especially problematic for motion refinement, where the goal is not to synthesize a new motion but to minimally refine an existing one.

MoreDiff addresses this limitation by reformulating motion refinement as manifold rectification. Instead of relying on long inpainting trajectories to accumulate correction, we train the diffusion model to learn a step-agnostic Manifold Rectification Gradient from q-sampled corrupted motions with clean-motion supervision. This enables effective few-step correction of structured mocap corruptions while preserving reliable regions of the original motion. In this sense, our method is complementary to prior diffusion motion models: rather than improving generation capacity alone, we adapt the diffusion prior specifically for high-fidelity motion cleanup.

3 Task Definition

This section defines the motion refinement task and introduces the diffusion-based manifold rectification view used in our method. We first describe the motion representation, and then formulate motion cleanup as pulling a corrupted motion sequence back toward the clean motion manifold while preserving its reliable regions.

3.1 Motion Representation

Let $x \in \mathbb{R}^{N \times D}$ denote a human motion sequence with N frames and feature dimension D. Each frame is represented as

$$\boldsymbol{\phi}_n = \left[\boldsymbol{\theta}_n, \boldsymbol{p}_n, \boldsymbol{r}_n\right], \tag{1}$$

where $\boldsymbol{\phi}_n \in \mathbb{R}^{201}$ is the motion state at frame n. The three components are:

- **Joint orientations** $\boldsymbol{\theta}_n \in \mathbb{R}^{132}$: global orientations of 22 joints represented in the continuous 6D rotation format;
- **Root-relative joint positions** $\boldsymbol{p}_n \in \mathbb{R}^{66}$: 3D positions of 22 joints relative to the root joint;
- **Global root translation** $\boldsymbol{r}_n \in \mathbb{R}^{3}$: the global translation of the root joint.

Thus, the feature dimension is $D = 132 + 66 + 3 = 201$. This representation combines orientation, body geometry, and global trajectory information, which are all important for preserving motion fidelity during refinement.

3.2 Motion Refinement as Manifold Rectification

We denote the set of clean, anatomically valid, and physically plausible human motions as a motion manifold $\mathcal{M}$. Given a corrupted motion sequence x_0^*, motion refinement aims to recover a clean motion $x_0 \in \mathcal{M}$ while preserving the reliable information in x_0^*:

$$\mathcal{R}(x_0^*) \longrightarrow x_0, \quad x_0 \in \mathcal{M}. \tag{2}$$

Unlike motion generation, x_0^* is not random noise but a mostly valid motion containing structured off-manifold corruptions, such as joint pops, pose twists, tracking drift, and foot penetration. Thus, refinement should correct corrupted regions without unnecessarily regenerating the entire motion.

We view diffusion-based refinement as **score-guided manifold rectification**. For a clean motion x_0, the standard forward diffusion process is

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}), \tag{3}$$

where t is the diffusion timestep and $\bar{\alpha}_t$ is the cumulative noise schedule. The learned score function $s_\theta(x_t, t) \approx \nabla_{x_t} \log p_t(x_t)$ provides the denoising direction toward high-density regions of the clean motion distribution. With noise prediction parameterization, it is written as

$$s_\theta(x_t, t) = -\frac{1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t). \tag{4}$$

Given a corrupted motion x_0^*, existing diffusion-based refinement methods first perturb it to a noisy state

$$x_t^* = q(x_0^*, t) = \sqrt{\bar{\alpha}_t} x_0^* + \sqrt{1-\bar{\alpha}_t}\epsilon, \tag{5}$$

and then use the reverse process to pull it toward the clean motion prior. Abstracting away sampler-specific details, this process can be interpreted as accumulating score-guided rectification:

$$\hat{x}_0 \approx q(x_0^*, T) + \sum_{t=1}^{T} c_t \cdot s_\theta(x_t^*, t), \tag{6}$$

where c_t is determined by the noise schedule and sampler. This formulation highlights our key observation: **the rectification strength of standard diffusion is inherently tied to the timestep** t.

Specifically, the standard score is mainly trained to remove Gaussian noise around clean samples, and provides only weak effective rectification for structured off-manifold corruptions:

$$\left\| s_\theta(x_t^*, t) \right\| \approx 0 \quad \text{if } t \to 0. \tag{7}$$

When T is large, the accumulated score guidance becomes strong enough to refine severe corruptions, but the heavily noised input loses more performance-specific details.

This timestep dependence explains the quality-fidelity dilemma. When T is small, x_t^* remains close to x_0^*, so the input fidelity is well preserved.

MoreDiff breaks this coupling by learning a **step-agnostic Manifold Rectification Gradient**. Given a clean motion x_0 and its corrupted counterpart x_0^*, we q-sample the corrupted motion:

$$x_t^* = \sqrt{\bar{\alpha}_t} x_0^* + \sqrt{1-\bar{\alpha}_t}\epsilon. \tag{8}$$

Instead of reconstructing x_0^*, the model is supervised to predict the clean counterpart x_0:

$$\mathcal{L}_{\text{rect}} = \mathbb{E}_{x_0, x_0^*, t, \epsilon} \left[\left\| f_\theta(x_t^*, t) - x_0 \right\|_2^2 \right]. \tag{9}$$

This objective explicitly teaches the model to map corrupted off-manifold states back to the clean motion manifold. The induced rectification signal is defined as

$$\boldsymbol{g}_\theta(x_t^*, t) = f_\theta(x_t^*, t) - x_0^*, \quad \left\| \boldsymbol{g}_\theta(x_t^*, t) \right\| \not\approx 0, \quad \forall t, \; x_0^* \notin \mathcal{M}. \tag{10}$$

Unlike the standard generation gradient that depends strongly on the diffusion step, $\boldsymbol{g}_\theta$ is directly supervised by corrupted-clean pairs across timesteps. Therefore, MoreDiff enabling few-step motion cleanup that improves quality while preserving the fidelity of the original performance.

4 Method

4.1 Overview

MoreDiff is designed to learn a step-agnostic Manifold Rectification Gradient (MRG) that directly maps structured off-manifold corruptions back toward the clean motion manifold. As shown in Fig. 3, the framework contains three components. First, an ad-hoc corruption synthesis pipeline generates paired corrupted-clean motions from clean training sequences. These pairs expose the model to common mocap corruptions and provide direct supervision for manifold rectification. Second, a clean-corrupt mixing training objective preserves the standard clean-motion diffusion prior while injecting MRG through corrupted-motion supervision. A geometric consistency loss further regularizes the predicted skeleton to maintain anatomically consistent offsets. Third, an adaptive rectification strategy (Ada-Rec) preserves regions that are already reliable and applies rectification only where the model detects corruption, reducing unnecessary changes to the original performance.

4.2 Ad-hoc Motion Corruption Synthesis

Learning MRG requires corrupted-clean pairs, but real mocap errors rarely come with manually corrected clean targets. We therefore synthesize structured corruptions from clean motions during training. The goal is not to exactly reproduce every capture failure, but to cover representative off-manifold deviations that commonly appear in optical, inertial, and video-based mocap.

Our corruption pipeline includes eight representative artifact types, such as joint pops, pose twists, tracking drift, frozen frames, foot-related physical violations, and IK-induced ambiguities. For each training motion, corruptions are applied with randomized temporal extents, affected joints, and intensities. We adopt a segmented random application strategy: each motion sequence is split into 10–30 frame segments, and each segment either remains clean or receives one corruption type. This randomized synthesis encourages the model to learn a general rectification behavior rather than overfitting to a fixed artifact pattern. Detailed corruption definitions and parameter ranges are provided in Appendix A.

4.3 Manifold Rectification Training

The model is built on a masked motion diffusion backbone. During training, we use a 50/50 clean-corrupt mixing strategy. For the clean half of each batch, the model follows standard diffusion training. Given a clean motion x_0, we sample

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \tag{11}$$

and train the model to reconstruct x_0. This branch maintains the clean-motion prior.

For the corrupted half, we first synthesize a corrupted counterpart x_0^* and apply the same q-sampling process:

$$x_t^* = \sqrt{\bar{\alpha}_t} x_0^* + \sqrt{1-\bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \tag{12}$$

Different from standard diffusion training, the supervision target remains the clean motion x_0, not the corrupted input x_0^*. This forces the network to learn a correction from q-sampled corrupted states to the clean manifold.

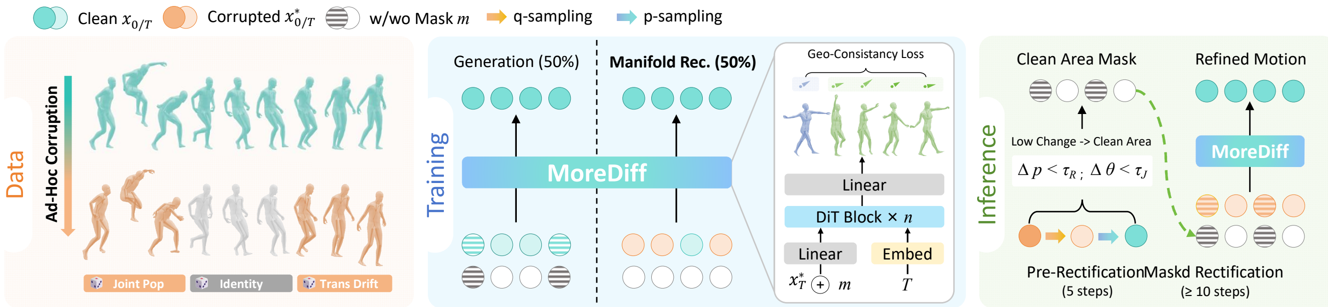

Fig. 3. Overview of MoreDiff. Left: we synthesize structured motion corruptions from clean motions to obtain corrupted-clean pairs. Middle: the diffusion model is trained with a clean-corrupt mixing objective, combining standard clean-motion diffusion training with manifold rectification training, augmented by a geometric consistency loss. Right: at inference time, Ada-Rec first estimates reliable regions through lightweight pre-rectification and then performs masked rectification, correcting corrupted regions while preserving reliable motion content.

The total diffusion objective is decomposed into two parts:

$$\mathcal{L}_{\text{gen}} = \frac{1}{N_{\text{clean}}} \sum_{i \in \mathcal{S}_{\text{clean}}} \left\| \hat{x}_0^{(i)} - x_0^{(i)} \right\|_1, \tag{13}$$

$$\mathcal{L}_{\text{MRG}} = \frac{1}{N_{\text{corrupt}}} \sum_{i \in \mathcal{S}_{\text{corrupt}}} \left\| \hat{x}_0^{(i)} - x_0^{(i)} \right\|_1. \tag{14}$$

Here $\mathcal{L}_{\text{gen}}$ preserves the basic denoising ability of the diffusion model, while $\mathcal{L}_{\text{MRG}}$ injects step-agnostic rectification signals into the model. Since corrupted samples are supervised by their clean counterparts across timesteps, the learned correction depends on the structured corruption rather than only on the diffusion noise.

4.3.1 Geometric Consistency Loss. Motion refinement should not only reduce numerical error, but also maintain the rigid structure of the human skeleton. We therefore introduce a self-supervised geometric consistency loss on the predicted clean motion. Given predicted global joint orientations R and root-relative joint positions $\boldsymbol{p}$, we compute each parent-child offset in the local frame of the parent joint:

$$\Delta_{p,c}(t) = R_p(t)^T \left(\boldsymbol{p}_c(t) - \boldsymbol{p}_p(t) \right), \tag{15}$$

where p and c denote a parent joint and its child. Since the skeleton bone structure should remain constant over time, we enforce the local offsets to be consistent with the first frame:

$$\mathcal{L}_{\text{consis}} = \left\| \Delta_{p,c}(t) - \Delta_{p,c}(0) \right\|_1. \tag{16}$$

For variable-length sequences, a temporal validity mask is applied so that padded frames do not contribute to the loss. To avoid destabilizing early training, $\mathcal{L}_{\text{consis}}$ is activated after a warm-up period:

$$\mathcal{L}_{\text{total}} = \begin{cases} \mathcal{L}_{\text{gen}} + \mathcal{L}_{\text{MRG}}, & \text{if iter} \le n_{\text{cs}}, \\ \mathcal{L}_{\text{gen}} + \mathcal{L}_{\text{MRG}} + \mathcal{L}_{\text{consis}}, & \text{otherwise.} \end{cases} \tag{17}$$

We set $n_{\text{cs}} = 10000$ in all experiments.

4.4 Adaptive Rectification Strategy (Ada-Rec)

Although MRG enables strong few-step correction, applying rectification uniformly to the whole sequence may still modify regions that are already reliable. To further preserve fidelity, we introduce *adaptive rectification* (Ada-Rec) at inference time. Ada-Rec first performs low step pre-rectification and compares the refined output with the input. Regions with small rotation and joint-position changes are treated as reliable and be preserved.

Algorithm 1 summarizes this procedure. We use a 5-step pre-rectification pass to obtain x_{pre}, compute joint rotation and position differences, and construct a keep mask M_{keep} using thresholds of $\tau_R = 5°$ and $\tau_J = 2$ cm. The final denoising pass then rectifies only the unmasked regions. This makes MoreDiff behave as a targeted cleanup tool: corrupted regions are pulled toward the clean manifold, while reliable motion content remains close to the original input.

ALGORITHM 1: Adaptive Motion Rectification

Require: Corrupted input motion x_{in}, total steps T, rotation threshold τ_R, joint position threshold τ_J,
Ensure: Refined motion x_{out}

```
% Pre-rectification with lightweight denoising
x_pre ← Denoise(x_in, steps = 5)
ΔR ← RotationDiff(x_in, x_pre)
ΔJ ← JointPosDiff(x_in, x_pre)
% Estimate clean preservation mask
M_R ← (ΔR < τ_R)
M_J ← (ΔJ < τ_J)
M_keep ← Concat(M_R, M_J)
% Final masked rectification
x_out ← Denoise(x_in, steps = T, keep_mask = M_keep)
return x_out
```

5 Experiment Settings

5.1 Datasets and Evaluation Protocol

We train MoreDiff with clean motions from five open-source motion sources: AMASS [Mahmood et al. 2019], MOYO [Tripathi et al. 2023], LARa [Niemann et al. 2020], FineDance [Li et al. 2023], and Weizmann [Krebs et al. 2021]. To avoid overlap with the main benchmark, we exclude the validation and test split used by StableMotion [Mu et al. 2025] from AMASS. All training motions are converted to our unified representation, and synthetic corruptions are applied on-the-fly to construct corrupted-clean pairs for manifold rectification.

For evaluation, we use three datasets with increasing realism and difficulty. **CorruptedAMASS** is our controlled benchmark for Level 1. It uses the 1,312 clean SMPL+H motion sequences from the StableMotion test split and generates paired corrupted motions using our corruption synthesis strategy. Since each corrupted motion has a known clean counterpart, this benchmark allows direct evaluation of both fidelity to the clean motion and artifact removal. **MoreDiff-test** is our real optical mocap benchmark for Level 2. It contains 255 corrupted motion samples selected from MOYO, LARa, FineDance, and Weizmann. These samples are first filtered by physical artifact indicators, including jitter, foot skating, and joint-pop statistics, and are then manually reviewed to ensure visible real capture corruptions. **PopDanceSet** [Luo et al. 2024] is used for Level 3. It contains 1,036 monocular video-based motion samples extracted from in-the-wild dance videos, where the corruptions are substantially more severe than optical mocap, especially in joint jitter and global translation drift.

The key difference among the three levels is the availability of clean reference motions. Level 1 has paired clean targets, so fidelity metrics are computed against the clean reference. Levels 2 and 3 do not have ground-truth clean motions; in these cases, fidelity metrics are computed against the corrupted input and should be interpreted as *input deviation* rather than absolute correctness. Therefore, for real corruptions, a good refinement method should not simply minimize deviation to the input, but should improve motion quality while keeping the deviation controlled.

5.2 Baselines

We compare MoreDiff with two representative diffusion-based refinement baselines. **StableMotion** [Mu et al. 2025] is the closest refinement-oriented baseline. It first estimates frame-level quality indicators and then adaptively repaints low-quality regions using a long diffusion chain, typically requiring 700–1000 denoising steps. **CondMDI-K** denotes a conventional diffusion inpainting baseline CondMDI [Cohan et al. 2024] using K denoising steps. Since the original CondMDI training data overlaps with the StableMotion test split, we retrain a leakage-free model using the official implementation for fair comparison. We report multiple step counts, including $K = 10, 100, 250, 500$, to expose the step-dependent quality-fidelity behavior of standard diffusion refinement.

5.3 Evaluation Metrics

We evaluate both **fidelity** and **quality**. Fidelity metrics measure how much the refined motion deviates from a reference sequence. When clean ground truth is available, the reference is the clean motion; otherwise, the reference is the corrupted input. We report 1) MPJPE (Mean Per Joint Position Error, cm), which measures root-aligned joint position error; 2) WA-MPJPE (World-Aligned MPJPE, cm), which includes global translation; 3) WA-MPJAE (World-Aligned Mean Per Joint Angle Error, degrees), which measures joint orientation error; 4) RTE (Root Trajectory Error, cm), which measures global root trajectory deviation; and 5) Accel (Acceleration Error, m/s^2), which measures acceleration deviation.

Quality metrics measure physically or perceptually undesirable corruptions, independent of a paired clean reference. We report 1) FF (Foot Floating, mm) to measure floating-foot corruptions, 2) FP (Foot Penetration, mm) and 3) FP Rate (%) for foot-ground penetration severity and frequency, and 4) FS (Foot Skating, mm) and 5) FS Rate (%) for contact sliding severity and frequency. We also report 6) Jitter (m/s^3) for high-frequency motion variation and 7) Pop Rate (%) for joint-pop events. Different from contact errors, Jitter and Pop Rate are not monotonically lower-is-better: excessively high values indicate corruptions, while excessively low values may indicate over-smoothing and loss of motion details or impact.

5.4 Training Settings

MoreDiff is trained on 4 GPUs (32G) with a total batch size of 512. We adopt the AdamW optimizer with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-4}. The model is trained for 600k iterations, which takes approximately 7 days.

6 Evaluation on Motion Refinement

To comprehensively validate the effectiveness, generalization ability, and practical applicability of MoreDiff for motion refinement, we design a three-level evaluation protocol covering synthetic, real-world optical, and real-world monocular video mocap corruptions. Each level is tailored to answer a specific research question, forming a hierarchical validation chain that verifies our method from fundamental feasibility to practical boundary.

6.1 Level 1: Synthesized Corruptions

Level 1 validates the fundamental feasibility of gradient-guided manifold rectification under a controlled setting. We evaluate on **CorruptedAMASS**, which uses the 1,312 clean SMPL+H sequences from the StableMotion test split as clean references and synthesizes their corrupted counterparts with our corruption pipeline. The synthesized corruptions include representative structured errors such as joint pops, pose twists, tracking drift, frozen frames, and foot-related physical violations.

The key advantage of this benchmark is that every corrupted sequence x_0^* has a paired clean motion $x_0 \in \mathcal{M}$. We can therefore directly measure whether a refinement method moves corrupted motions toward the clean reference, rather than merely making the output smoother. This level tests whether MoreDiff can provide effective rectification while avoiding unnecessary changes to reliable motion content.

Table 1 shows that MoreDiff achieves the best fidelity to the clean reference across all reported error metrics. Compared with the corrupted input, MoreDiff reduces MPJPE from 2.78 cm to 1.05 cm, WA-MPJAE from 7.29° to 3.04°, and RTE from 15.21 cm to 5.19 cm. This confirms that the learned rectification direction moves corrupted motions toward their clean counterparts rather than merely producing smoother outputs. In contrast, StableMotion introduces larger deviations than the corrupted input on several fidelity metrics, indicating that long repainting can overwrite reliable motion content.

The quality metrics further reveal the role of few-step rectification. MoreDiff substantially reduces floor penetration and foot skating while keeping Jitter and Pop Rate close to the clean statistics. Conventional diffusion refinement shows the expected step-dependent

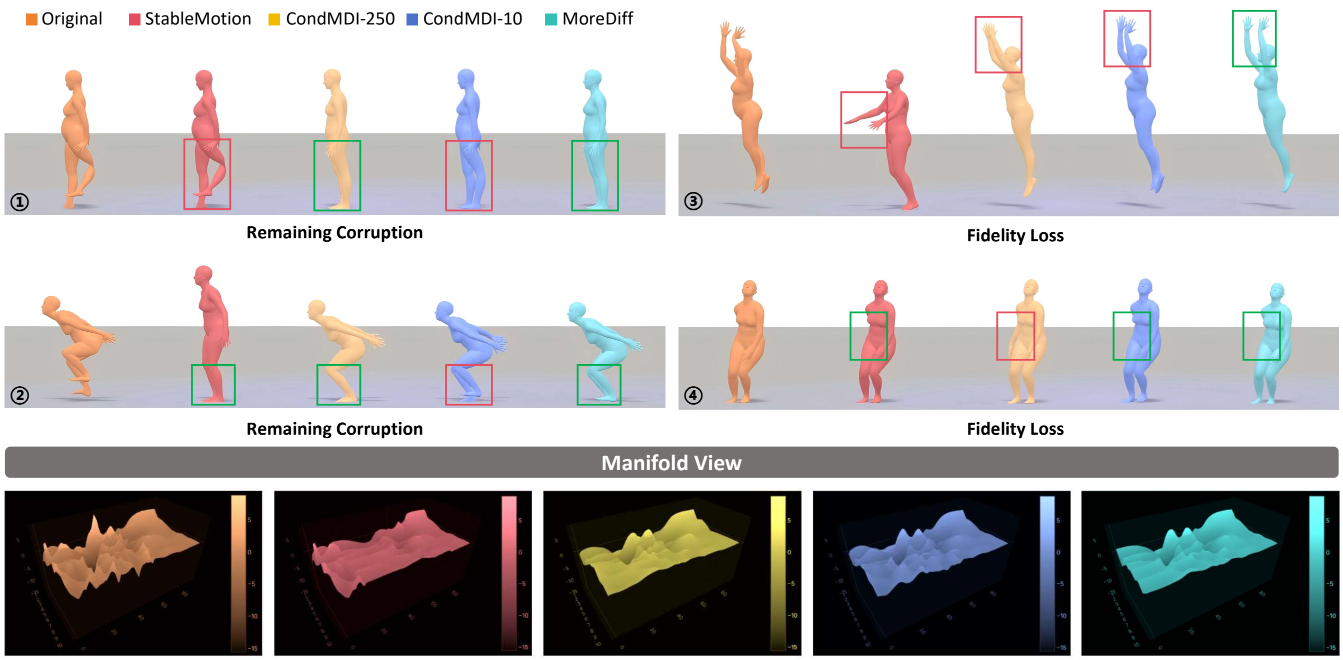

Fig. 4. Qualitative comparison on synthesized corruptions. StableMotion often leaves visible corruptions or changes reliable body regions, and conventional diffusion refinement shows a step-dependent trade-off between remaining corruption and fidelity loss. MoreDiff corrects local corruptions while better preserving the original pose and motion structure. The PCA-based manifold views at the bottom further illustrate that MoreDiff restores a smoother motion manifold without over-smoothing the input-specific motion pattern.

Table 1. Quantitative results on CorruptedAMASS (Level 1). Clean ground truth is available, so fidelity metrics are computed against the clean reference. ↓ means lower is better; ⟶ means closer to the clean motion statistics is preferred.

Method	Fidelity Metrics					Quality Metrics						
	MPJPE↓	WA-MPJPE↓	WA-MPJAE↓	RTE↓	Accel↓	FF↓	FP↓	FP Rate↓	FS↓	FS Rate↓	Jitter⟶	Pop Rate⟶
Clean Reference	/	/	/	/	/	6.99	10.25	5.91	9.65	13.18	0.07	0.09
Corrupted Input	2.78	17.34	7.29	15.21	3.96	7.13	63.13	14.68	14.40	13.69	2.20	7.58
StableMotion	$5.01^{\pm 4.47}$	$28.00^{\pm 36.79}$	$15.10^{\pm 14.44}$	$25.98^{\pm 36.86}$	$2.62^{\pm 3.45}$	7.16	68.17	19.83	16.08	16.22	1.19	2.33
CondMDI-500	$3.14^{\pm 2.07}$	$14.24^{\pm 17.81}$	$8.38^{\pm 6.49}$	$13.53^{\pm 17.75}$	$0.28^{\pm 0.31}$	6.80	14.70	14.73	9.93	14.27	0.06	0.04
CondMDI-250	$2.51^{\pm 1.73}$	$11.92^{\pm 16.10}$	$6.79^{\pm 6.16}$	$11.36^{\pm 16.02}$	$0.26^{\pm 0.27}$	6.97	14.40	13.92	10.39	14.68	0.06	0.04
CondMDI-100	$2.24^{\pm 1.67}$	$11.98^{\pm 17.44}$	$6.03^{\pm 5.86}$	$11.43^{\pm 17.38}$	$0.27^{\pm 0.27}$	7.36	14.62	13.01	11.00	15.62	0.07	0.08
CondMDI-10	$2.18^{\pm 1.88}$	$14.00^{\pm 21.26}$	$5.70^{\pm 5.40}$	$13.28^{\pm 21.26}$	$0.66^{\pm 0.88}$	8.43	20.71	12.62	13.31	16.36	0.26	1.23
MoreDiff (T=10)	$1.05^{\pm 1.31}$	$5.48^{\pm 13.43}$	$3.04^{\pm 4.50}$	$5.19^{\pm 13.29}$	$0.19^{\pm 0.20}$	7.47	**10.60**	**5.39**	**9.73**	**13.40**	0.06	0.07

behavior: small-step CondMDI preserves more input details but leaves stronger contact corruptions, while large-step CondMDI reduces some corruptions but suppresses temporal dynamics toward overly smooth values. These results directly support our central claim that step-agnostic MRG provides stronger refine without requiring long denoising trajectories that damage fidelity.

6.2 Level 2: Optical Mocap Corruptions

Level 2 evaluates synthetic-to-real generalization on real optical mocap corruptions. We created **MoreDiff-test**, a benchmark containing 255 corrupted samples selected from MOYO, LARa, FineDance, and Weizmann. These sequences come from high-accuracy optical capture sources, but still contain visible corruptions caused by capture noise, marker occlusion, or downstream pose fitting errors. We obtain the benchmark through physical artifact indicators filtering, such as jitter, foot skating, and joint-pop statistics, followed by manual review to confirm visually apparent corruption.

MoreDiff-test does not provide paired clean ground truth. Therefore, fidelity metrics in this level measure deviation from the corrupted input rather than absolute correctness. The goal is to test whether a model trained only with synthetic corruptions can improve real optical mocap quality while preserving the original captured performance.

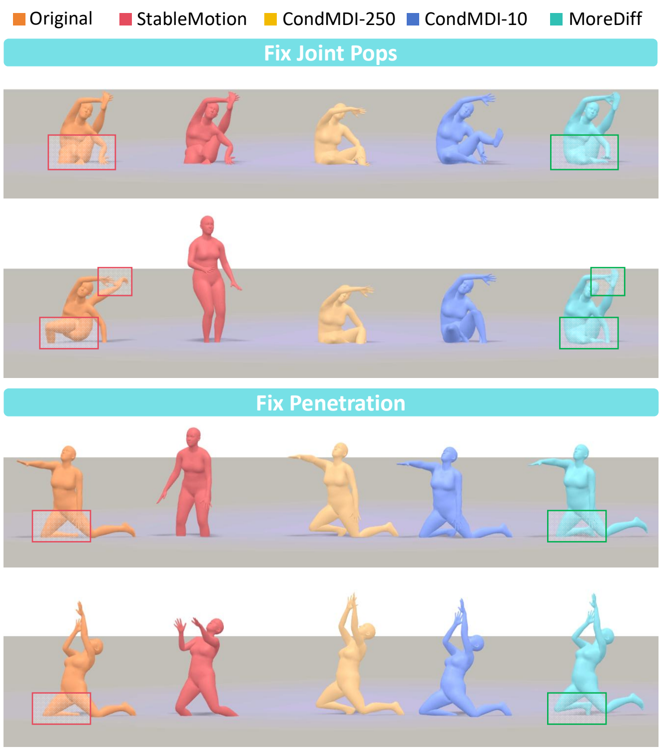

Fig. 5. Qualitative comparison on real optical mocap corruptions. MoreDiff corrects typical capture corruptions such as joint pops and penetration while preserving the original motion context.

As shown in Table 2, MoreDiff generalizes well to real optical mocap corruptions despite being trained only with synthetic corruptions. It achieves the smallest input deviation among all refinement methods on the fidelity metrics, while also reducing the main contact corruptions: floor penetration decreases from 12.19 mm to 6.97 mm, floor penetration rate from 13.15% to 10.25%, and foot skating from 13.10 mm to 12.25 mm. This behavior is desirable for real mocap cleanup, where the method should correct visible corruptions without drifting far from the captured performance.

The comparison also highlights the limitations of standard diffusion inpainting. Increasing the denoising length in CondMDI often produces stronger smoothing and larger deviation from the input, while shorter denoising better preserves the input but is less effective on contact corruptions. StableMotion can reduce some temporal corruptions, but its much larger pose and trajectory deviation suggests that quality-indicator-based repainting may modify reliable regions. MoreDiff provides a better balance: it removes physically implausible corruptions while maintaining controlled deviation, indicating that the rectification behavior learned from synthetic corruptions transfers to real optical mocap errors.

6.3 Level 3: Video Mocap Corruptions

Level 3 evaluates MoreDiff using **PopDanceSet**, which contains 1,036 motion samples extracted from dancing videos and exhibits stronger corruptions than optical mocap, especially high-frequency joint jitter and unstable global translation. Since no paired clean reference is available, this level tests whether MoreDiff can improve physical and temporal quality while maintaining controlled deviation from the original video-mocap estimate.

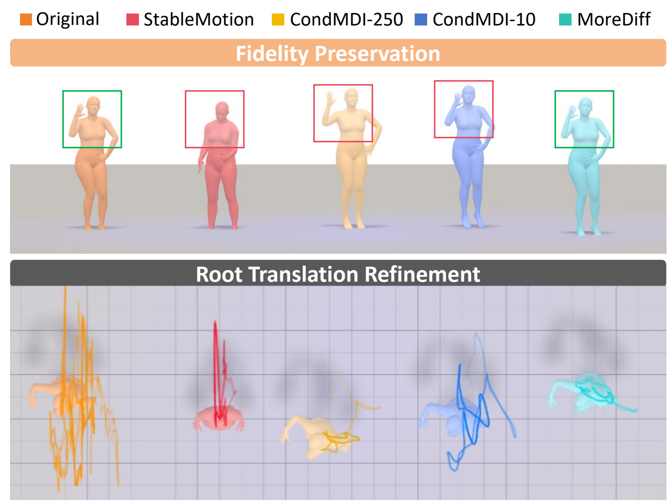

Fig. 6. Qualitative comparison on monocular video mocap corruptions. MoreDiff preserves input-specific pose details while refining unstable root translation. The trajectory visualization shows that MoreDiff reduces translation noise without collapsing the original motion path, demonstrating its ability to handle challenging video-based mocap corruptions.

Table 3 shows that video-based mocap contains substantially more severe artifacts than optical mocap, including foot skating, floor penetration, high jitter, and translation instability. MoreDiff achieves the lowest local pose deviation, with the best MPJPE and WA-MPJAE, while reducing several dominant physical artifacts such as FF, FS, and FS Rate compared with the corrupted input. The qualitative results in Fig. 6 further show that MoreDiff stabilizes the motion while preserving recognizable pose details from the input video estimate.

Because PopDanceSet has no clean reference, global fidelity metrics such as WA-MPJPE and RTE measure deviation from the noisy input trajectory rather than error to ground truth. MoreDiff may introduce larger global deviation than short-step CondMDI because it actively corrects unstable translation drift, while methods with smaller deviation may simply preserve the corrupted trajectory. Thus, Level 3 should be viewed as a practical stress test: MoreDiff improves local pose quality and reduces major physical artifacts, but does not claim to fully solve all severe video-mocap failures.

7 Ablation Study

We conduct ablations on CorruptedAMASS to evaluate the main components of MoreDiff: Manifold Rectification Gradient (MRG), geometric consistency loss $\mathcal{L}_{\text{consis}}$, and adaptive rectification (AdaRec). All variants use 10-step rectification and are progressively built from a masked motion model trained with the standard clean-motion diffusion objective.

Table 2. Quantitative results on MoreDiff-test (Level 2). Since no clean reference is available, fidelity metrics are computed against the corrupted input and represent input deviation; quality metrics report absolute artifact severity. For contact metrics, ↓ means lower is better; for Jitter and Pop Rate, → indicates that overly large values suggest corruptions while overly small values may indicate over-smoothing.

Method	Fidelity Metrics (vs Corrupted)					Quality Metrics						
	MPJPE↓	WA-MPJPE↓	WA-MPJAE↓	RTE↓	Accel↓	FF↓	FP↓	FP Rate↓	FS↓	FS Rate↓	Jitter→	Pop Rate→
Corrupted Input	/	/	/	/	/	8.76	12.19	13.15	13.10	15.29	0.50	3.46
StableMotion	$8.09^{\pm 5.45}$	$21.54^{\pm 21.48}$	$24.68^{\pm 16.33}$	$18.65^{\pm 21.97}$	$2.42^{\pm 1.77}$	9.90	20.90	15.85	15.38	16.76	0.86	1.42
CondMDI-500	$8.76^{\pm 3.97}$	$35.46^{\pm 17.76}$	$22.50^{\pm 7.83}$	$33.29^{\pm 19.05}$	$1.19^{\pm 0.96}$	10.11	18.30	27.17	12.85	20.09	0.10	0.10
CondMDI-250	$6.25^{\pm 3.33}$	$22.79^{\pm 10.41}$	$17.33^{\pm 7.78}$	$20.95^{\pm 11.26}$	$1.15^{\pm 0.95}$	9.96	17.08	25.51	13.87	20.95	0.11	0.16
CondMDI-100	$4.61^{\pm 2.91}$	$14.29^{\pm 6.54}$	$14.03^{\pm 7.47}$	$12.65^{\pm 6.81}$	$1.11^{\pm 0.95}$	10.15	14.96	20.89	14.12	19.72	0.13	0.24
CondMDI-10	$2.77^{\pm 1.92}$	$7.91^{\pm 5.72}$	$9.87^{\pm 5.85}$	$6.76^{\pm 5.72}$	$0.99^{\pm 0.93}$	10.67	10.16	14.62	13.95	16.50	0.17	0.54
MoreDiff (T=10)	$1.55^{\pm 1.12}$	$7.03^{\pm 13.11}$	$5.51^{\pm 4.08}$	$6.55^{\pm 12.98}$	$0.92^{\pm 0.97}$	10.35	**6.97**	**10.25**	**12.25**	**14.64**	0.11	0.43

Table 3. Quantitative results on PopDanceSet (Level 3). Since no clean reference is available, fidelity metrics are computed against the corrupted input and represent input deviation; quality metrics report absolute artifact severity. For contact metrics, ↓ means lower is better; for Jitter and Pop Rate, → indicates that overly large values suggest artifacts while overly small values may indicate over-smoothing.

Method	Fidelity Metrics (vs Corrupted)					Quality Metrics						
	MPJPE↓	WA-MPJPE↓	WA-MPJAE↓	RTE↓	Accel↓	FF↓	FP↓	FP Rate↓	FS↓	FS Rate↓	Jitter→	Pop Rate→
Corrupted Input	/	/	/	/	/	10.25	37.72	19.00	155.53	34.39	15.91	6.15
StableMotion	$13.25^{\pm 3.40}$	$52.26^{\pm 26.87}$	$47.70^{\pm 12.84}$	$48.23^{\pm 27.09}$	$29.20^{\pm 11.72}$	8.50	60.62	57.50	13.16	8.90	1.53	1.22
CondMDI-500	$7.21^{\pm 1.40}$	$50.15^{\pm 27.48}$	$20.88^{\pm 3.15}$	$49.26^{\pm 27.53}$	$29.77^{\pm 12.22}$	5.63	31.57	55.94	11.81	28.83	0.13	0.07
CondMDI-250	$4.91^{\pm 1.02}$	$43.52^{\pm 24.34}$	$15.42^{\pm 2.89}$	$43.01^{\pm 24.35}$	$29.75^{\pm 12.22}$	6.71	26.01	47.20	15.01	36.42	0.17	0.22
CondMDI-100	$3.32^{\pm 0.85}$	$38.30^{\pm 22.06}$	$12.18^{\pm 2.70}$	$37.97^{\pm 22.04}$	$29.71^{\pm 12.22}$	8.28	17.06	32.67	16.98	37.58	0.21	0.56
CondMDI-10	$1.93^{\pm 0.64}$	$29.86^{\pm 18.19}$	$8.87^{\pm 2.26}$	$29.68^{\pm 18.18}$	$29.52^{\pm 12.20}$	11.14	11.47	14.52	24.15	35.93	0.43	1.59
MoreDiff (T=10)	$1.70^{\pm 1.02}$	$51.42^{\pm 37.92}$	$7.40^{\pm 3.66}$	$51.33^{\pm 37.89}$	$29.61^{\pm 12.22}$	6.78	17.57	34.87	14.94	32.83	0.35	1.68

Table 4. Ablation study on key components for motion fidelity and motion quality on CorruptedAMASS. We set T=10 for all ablation cases. ↓ means lower is better; → means closer to the clean motion statistics is preferred.

Method	Fidelity Metrics					Quality Metrics						
	MPJPE↓	WA-MPJPE↓	WA-MPJAE↓	RTE↓	Accel↓	FF↓	FP↓	FP Rate↓	FS↓	FS Rate↓	Jitter→	Pop Rate→
Clean	/	/	/	/	/	6.99	10.25	5.91	9.65	13.18	0.07	0.09
Input	$2.74^{\pm 3.16}$	$31.93^{\pm 46.16}$	$9.16^{\pm 7.25}$	$30.02^{\pm 46.78}$	$4.15^{\pm 4.97}$	7.13	63.13	14.68	14.40	13.69	2.20	7.58
Base MMM	$2.41^{\pm 2.34}$	$16.04^{\pm 24.00}$	$6.58^{\pm 7.53}$	$15.16^{\pm 23.98}$	$0.97^{\pm 1.32}$	7.78	35.72	14.45	13.69	14.97	0.42	2.89
+MRG	$1.43^{\pm 1.04}$	$6.42^{\pm 9.75}$	$4.02^{\pm 3.22}$	$6.06^{\pm 9.72}$	$0.22^{\pm 0.21}$	6.26	12.07	11.77	10.40	14.94	0.07	0.06
+MRG $+\mathcal{L}_{\text{consis}}$	$1.27^{\pm 1.27}$	$5.78^{\pm 11.04}$	$3.59^{\pm 4.38}$	$5.45^{\pm 10.93}$	$0.21^{\pm 0.21}$	5.77	16.36	15.02	7.15	6.98	0.07	0.07
+MRG $+\mathcal{L}_{\text{consis}}$ *+Ada-Rec*	$\mathbf{1.05}^{\pm 1.31}$	$\mathbf{5.48}^{\pm 13.43}$	$\mathbf{3.04}^{\pm 4.50}$	$\mathbf{5.19}^{\pm 13.29}$	$\mathbf{0.19}^{\pm 0.20}$	7.47	10.60	5.39	9.73	13.40	0.06	0.07

Table 4 shows that MRG provides the primary refinement capability, substantially reducing pose, orientation, trajectory, and penetration errors compared with Base MMM. This confirms that corrupted-to-clean supervision learns a direct rectification signal rather than merely smoothing the motion. $\mathcal{L}_{\text{consis}}$ further improves skeletal coherence and foot-skating behavior, while Ada-Rec achieves the best fidelity by preserving reliable regions and suppressing over-rectification. Together, these results validate the three key designs of MoreDiff and show that they jointly improve motion quality while maintaining fidelity to the clean reference.

7.1 Validation on Manifold Rectification Training

We further analyze whether Manifold Rectification Training changes the direction and stability of the score-guided update. For a corrupted input x_0^*, its paired clean motion x_0, and a refined result $\hat{x}_0$, we define the refinement displacement as $\mathbf{v}_{\text{fix}} = \hat{x}_0 - x_0^*$ and the target clean direction as $\mathbf{v}_{\text{clean}} = x_0 - x_0^*$. The effective rectification component along the clean direction is measured by

$$\text{Proj}(\mathbf{v}_{\text{fix}}, \mathbf{v}_{\text{clean}}) = \frac{\mathbf{v}_{\text{fix}} \cdot \mathbf{v}_{\text{clean}}}{\|\mathbf{v}_{\text{clean}}\|}. \tag{18}$$

A larger positive value means that the update contains a stronger component toward the clean counterpart, while a value that grows only at large steps indicates timestep-dependent rectification.

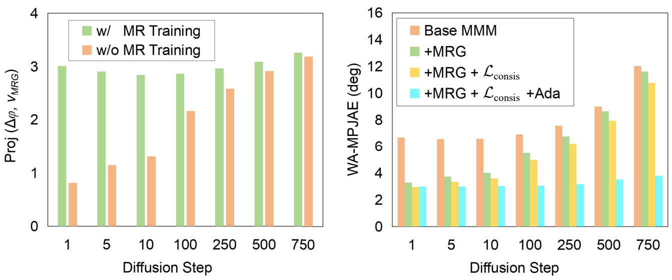

Fig. 7. Ablation analysis of the proposed rectification design. Left: Manifold Rectification Training provides a stable rectification component along the clean-manifold direction across diffusion steps, whereas the baseline MMM obtains strong correction only when the diffusion step becomes large. Right: adding MRG, geometric consistency, and Ada-Rec progressively reduces WA-MPJAE, especially under large denoising steps where standard diffusion refinement suffers from fidelity degradation.

As shown in Fig. 7 (left), the baseline MMM exhibits a strongly step-dependent behavior: its rectification component is weak at small steps and increases mainly when the diffusion process becomes long. This matches the quality-fidelity dilemma discussed earlier. Strong correction requires large denoising steps, but such steps also erase more input information. In contrast, the model trained with MRG maintains a stable positive rectification component across diffusion steps. This supports our central claim that MoreDiff decouples rectification strength from denoising length.

Fig. 7 (right) further shows that the full design is most robust when the denoising step increases. Base MMM and the variants without Ada-Rec suffer from larger WA-MPJAE at high steps, indicating fidelity degradation caused by over-rectification. The full model maintains low angular error across the tested steps, confirming all the design contribute to the final performance.

7.2 Validation on $\mathcal{L}_{consis}$

The geometric consistency loss is designed to preserve the rigid skeletal structure implied by the input sequence. Its effect is visible in both fidelity and physical quality. In Table 4, adding $\mathcal{L}_{consis}$ on top of MRG reduces MPJPE, WA-MPJPE, WA-MPJAE, and RTE, showing that the refined poses become closer to the clean reference. At the same time, it substantially improves foot-skating metrics. This suggests that geometric consistency does not simply regularize each pose independently; it also reduces ambiguity between local joint rotations and global translation, leading to more coherent body-ground motion.

7.3 Validation on Ada-Rec Strategy

Ada-Rec is a fidelity-preserving inference strategy. It first performs a lightweight pre-rectification pass to identify joints and frames that are already reliable, and then preserves these regions during the final rectification. This design prevents the model from treating the entire sequence as corrupted and makes MoreDiff behave more like a targeted cleanup tool.

The ablation results show that Ada-Rec improves the full model primarily by reducing unnecessary deviation. Compared with the variant using MRG and $\mathcal{L}_{consis}$, Ada-Rec further lowers MPJPE from 1.27 cm to 1.05 cm, WA-MPJAE from 3.59° to 3.04°, and RTE from 5.45 cm to 5.19 cm. It also reduces floor penetration from 16.36 mm to 10.60 mm and floor penetration rate from 15.02% to 5.39%. Although foot-skating metrics become slightly higher than the variant without Ada-Rec, they remain close to the clean reference and much better than the corrupted input. This behavior is consistent with the role of Ada-Rec: it does not maximize smoothing or physical correction at any cost, but preserves reliable motion while allowing corrupted regions to be rectified.

8 Applications

Mocap Data Refinement. We evaluate MoreDiff on the TotalCapture dataset, which contains 45 optical motion capture sequences. Since TotalCapture does not provide paired clean ground-truth motions, the goal is not to minimize all deviations, but to improve motion quality while preserving the captured performance.

Table 5 shows that MoreDiff achieves a better quality-fidelity trade-off than existing methods. StableMotion can reduce some artifacts, but its long-step repainting often causes excessive deviation from the input, especially when corrupted regions are not accurately localized by its quality indicators. CondMDI with few denoising steps better preserves the input but provides limited correction, while using more steps increases the risk of over-smoothing. In contrast, MoreDiff applies targeted rectification and maintains controlled input deviation, demonstrating that our synthetic-corruption training can generalize to real optical mocap refinement without requiring clean ground-truth supervision.

8.1 Acquiring Animation Assets from Video Generation

We further evaluate MoreDiff on a small generated-video test set. We collected 122 human-motion videos from video generation, covering sports, daily activities, dance, and stage performance. We employed GVHMR [Shen et al. 2024] to extract SMPL-X motion sequences from these generated videos, and then apply MoreDiff for motion refinement.

Since no clean reference is available, fidelity metrics are computed against the original captured motion from GVHMR. The goal is therefore reduce physical corruptions while preserving the intended motion visible in the video. Fig. 8 illustrates this process: MoreDiff preserves the overall pose and timing while correcting foot floating and skating..

Table 6 confirms the qualitative observation. MoreDiff produces the smallest local deviation from the input motion, with the lowest MPJPE (1.22 cm) and WA-MPJAE (3.11°), while substantially improving contact quality. It reduces floor penetration from 23.62 mm to 4.43 mm and the floor-penetration rate from 16.55% to 6.91%. The RTE is higher than CondMDI-10 because MoreDiff actively changes unstable global translation when correcting contacts. The generated-video setting also exposes the limitation of long-chain inpainting for high-noise video mocap. StableMotion worsens several contact metrics and increases Jitter from 0.19 to 0.69, indicating that its quality-indicator based repainting might be unreliable when video mocap corruptions are severe and spatially widespread. CondMDI with large steps can reduce floor penetration, but introduces

Table 5. Quantitative results on the TotalCapture mocap dataset. Since ground-truth poses are available, fidelity metrics are computed against the ground truth. ↓ means lower is better; → means closer to the clean/reference motion statistics is preferred for Jitter and Pop Rate.

Method	Fidelity Metrics				Quality Metrics						
	MPJPE↓	WA-MPJPE↓	WA-MPJAE↓	RTE↓	FF↓	FP↓	FP Rate↓	FS↓	FS Rate↓	Jitter→	Pop Rate→
GT (Corrupted)	-	-	-	-	9.71	5.09	2.39	10.95	17.88	0.095	0.11
StableMotion	$3.09^{\pm 2.47}$	$12.40^{\pm 10.03}$	$9.00^{\pm 6.45}$	$11.29^{\pm 9.83}$	10.37	5.32	2.33	15.79	20.87	0.496	0.15
CondMDI-250	$1.20^{\pm 0.90}$	$30.37^{\pm 15.40}$	$4.97^{\pm 2.31}$	$30.28^{\pm 15.41}$	**9.67**	9.36	19.42	12.58	11.50	0.079	0.03
CondMDI-100	$1.08^{\pm 0.78}$	$18.07^{\pm 10.51}$	$4.60^{\pm 2.05}$	$17.97^{\pm 10.53}$	10.68	5.58	9.09	11.08	7.52	0.070	0.02
CondMDI-10	$0.87^{\pm 0.56}$	$11.80^{\pm 6.14}$	$3.91^{\pm 1.53}$	$11.71^{\pm 6.14}$	10.43	**3.07**	3.61	**10.36**	**4.59**	0.063	0.03
MoreDiff (T=10)	$0.72^{\pm 0.34}$	$14.11^{\pm 12.00}$	$2.64^{\pm 1.03}$	$14.06^{\pm 11.99}$	9.94	3.93	6.72	10.92	11.24	**0.056**	0.03

Table 6. Quantitative results on the generated-video GVHMR test set. Since no clean reference is available, fidelity metrics are computed against the corrupted input and represent input deviation; quality metrics report absolute artifact severity. For contact metrics, ↓ means lower is better; for Jitter and Pop Rate, → indicates that overly large values suggest corruptions while overly small values may indicate over-smoothing.

Method	Fidelity Metrics (vs Corrupted)					Quality Metrics						
	MPJPE↓	WA-MPJPE↓	WA-MPJAE↓	RTE↓	Accel↓	FF↓	FP↓	FP Rate↓	FS↓	FS Rate↓	Jitter↓	Pop Rate↓
Corrupted Input	/	/	/	/	/	8.16	23.62	16.55	12.02	13.82	0.19	0.13
StableMotion	$3.04^{\pm 6.36}$	$7.86^{\pm 25.27}$	$7.59^{\pm 13.54}$	$6.24^{\pm 24.55}$	$1.65^{\pm 5.83}$	**8.47**	26.38	17.58	13.98	**14.15**	0.69	0.24
CondMDI-250	$6.35^{\pm 3.22}$	$25.55^{\pm 21.17}$	$14.07^{\pm 5.50}$	$24.13^{\pm 21.24}$	$0.72^{\pm 0.91}$	10.33	7.93	11.49	**10.53**	15.21	**0.113**	0.15
CondMDI-100	$4.08^{\pm 2.51}$	$15.58^{\pm 13.07}$	$9.68^{\pm 4.54}$	$14.70^{\pm 12.90}$	$0.67^{\pm 0.92}$	9.91	**7.40**	10.94	11.40	17.31	0.129	0.16
CondMDI-10	$1.81^{\pm 1.41}$	$7.43^{\pm 7.89}$	$4.92^{\pm 2.78}$	$7.03^{\pm 7.79}$	$0.53^{\pm 0.78}$	9.44	9.48	10.81	11.67	16.85	0.136	0.15
MoreDiff (T=10)	$1.22^{\pm 1.79}$	$12.40^{\pm 25.97}$	$3.11^{\pm 3.01}$	$12.23^{\pm 25.92}$	$0.47^{\pm 0.82}$	8.94	**4.43**	**6.91**	10.70	15.68	0.116	0.17

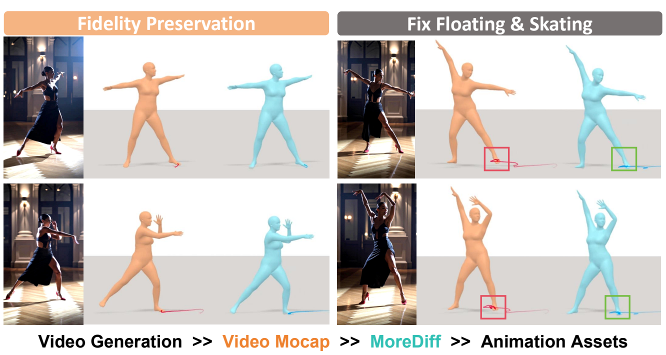

Fig. 8. Application on animation asset acquisition from generated videos.

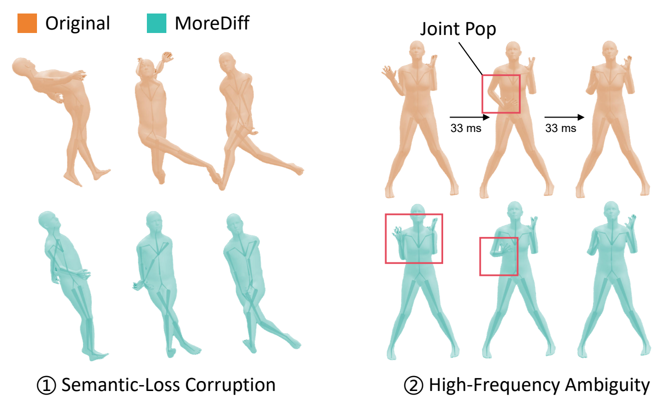

Fig. 9. Typical failure cases of MoreDiff. Left: semantic-loss corruption. Right: high-frequency ambiguity.

pose and translation deviation. While MoreDiff removes major foot-contact corruptions while keeping the refined motion close enough to the video-derived performance for downstream editing.

9 Limitation and Discussion

Although MoreDiff performs well in common motion refinement scenarios, it has limitations under extreme corruption and complex motion patterns, with two representative failure cases (Fig. 9).

First, MoreDiff fails under severe **Semantic Ambiguity**, such as left-right inversion, body-part confusion, or heavily broken configurations, where original motion semantics are largely lost. Since manifold rectification relies on valid input information to guide motion back to the clean manifold, it becomes ineffective without such semantic cues. Here, the refinement task degenerates into full motion reconstruction, where inpainting/generation-based paradigms are more suitable—clarifying the two paradigms' complementarity: rectification for semantically meaningful inputs, and inpainting for inputs lacking reliable semantics.

Second, MoreDiff struggles with **Spectral Ambiguity** in highly dynamic motions. Fast actions contain legitimate high-frequency signals that are hard to distinguish from jitter-like corruption, leading to residual jitter or over-suppressed valid details. This indicates limited ability to separate motion detail from high-frequency noise in extreme dynamic scenes.

While effective for common structured mocap corruptions, MoreDiff still faces challenges in extreme semantic loss and high-frequency ambiguity. Future work will expand synthetic corruption space to cover edge cases and introduce frequency-aware modeling to distinguish valid dynamic details from jitter.

10 Conclusion

We proposed MoreDiff, a diffusion-based motion refinement framework that formulates motion cleanup as manifold rectification. Conventional diffusion refinement suffers from a quality-fidelity dilemma due to coupled rectification strength and denoising length; MoreDiff breaks this coupling by learning a step-agnostic Manifold Rectification Gradient from q-sampled corrupted motions with clean-motion supervision.

Integrating synthetic corruption training, geometric consistency regularization, and adaptive rectification, MoreDiff achieves targeted few-step cleanup while preserving reliable motion regions. Experiments across synthetic, real optical mocap, monocular video mocap, and video-to-animation applications show MoreDiff improves motion quality with better fidelity than inpainting-style diffusion refinement. Critically, trained only on synthetic corruptions, it generalizes to real capture corruptions, providing a low-cost solution for motion capture post-processing and animation cleanup.

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A Motion Corruption Synthesis Settings

Our corruption patterns are derived from two primary categories of real-world mocap system errors:

- **Optical Mocap Errors**: Marker confusion, occlusion, triangulation noise, and temporary marker loss.
- **Video Mocap Errors**: 2D keypoint detection noise, depth ambiguity, view-dependent distortion, and inverse kinematics (IK) multi-solution ambiguity.

By simulating these errors with controlled parameters, we provide diverse samples that covers common motion corruptions in practical applications.

A.1 Corruption Patterns

We implement eight representative corruption types, each with configurable density (ratio of affected joints/frames) and scale (degradation magnitude) parameters:

Joint Orientation/Rotation Pops Simulates sudden changes in joint orientation caused by marker confusion (optical mocap) or 2D keypoint detection noise (video mocap). We randomly select a subset of joints and timesteps, apply random Euler angle perturbations (up to 180°), and optionally smooth the perturbation across adjacent frames to simulate temporal correlation. Two variants are provided: - *Orientation pops*: Directly modify local joint rotation matrices - *Rotation pops*: Apply perturbations through inverse kinematics to preserve plausible joint positions

Pose Twist Models continuous pose distortion caused by body self-occlusion or partial field-of-view loss. Unlike sudden pops, this corruption applies temporally smooth rotational perturbations to a subset of joints, with root joint perturbations scaled down by a factor of 3 to maintain global motion consistency. Perturbations are zeroed at the beginning and end of segments to simulate gradual onset/offset of occlusion.

Candy Wrapper Twist Addresses the IK multi-solution problem where multiple joint rotation combinations produce identical end-effector positions. We target pre-defined IK-ambiguous joints (shoulders, hips, elbows, knees) and apply random rotations around the axis defined by the joint and its child, while keeping the child joint position unchanged. This simulates the "candy wrapper" effect where joint orientation flips to a mathematically valid but anatomically incorrect solution.

Frozen Frame Simulates complete tracking failure when the subject exits the camera field of view. We randomly select a continuous segment of frames and replace all frames in the segment with the last valid frame before the segment, mimicking the behavior of real mocap systems that repeat the last known pose during tracking loss.

Translation Drift Models the cumulative effect of camera position estimation errors in monocular video mocap. We apply a combination of linear drift and Gaussian random walk to the root joint translation increments, with optional temporal smoothing using 3-7 frame sliding windows. The drift accumulates over time but does not affect relative joint positions.

Translation Distortion Simulates depth estimation errors and camera pose misalignment. We apply independent scaling factors (0.1-1.5×) to the X, Y, Z axes of root translation increments, with larger distortion typically applied to the depth axis. We also add random Yaw rotation (-30° to 30°) to simulate global orientation misalignment.

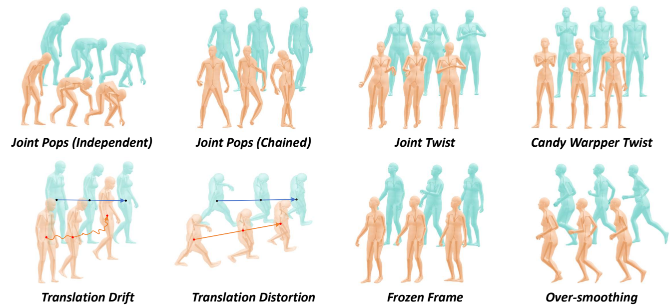

Fig. 10. The 8 Core Corruption Patterns used in MoreDiff Training.

Table 7. Motion Corruption Parameter Configuration

Corruption Type	Density Range	Scale Range	Probability
Identity	/	/	33.3%
Joint Orientation Pops	[0.05, 0.50]	[0.1, 1.0]	8.3%
Joint Rotation Pops	[0.05, 0.20]	[0.1, 0.5]	8.3%
Pose Twist	[0.05, 0.20]	[0.1, 1.0]	8.3%
Candy Wrapper Twist	[0.05, 0.50]	[0.25, 1.0]	8.3%
Frozen Frame	[0.1, 0.5]	/	8.3%
Translation Drift	[0.1, 1.0]	[0.1, 0.3]	8.3%
Translation Distortion	/	[0.1, 1.0]	8.3%
Over-Smooth	[0.1, 0.5]	[0.0, 0.0]	8.3%

Over-Smooth Models excessive post-processing or over-aggressive filtering in video mocap systems. We randomly select a subset of joints and apply random temporal smoothing to their rotation sequences using variable kernel sizes (3, 5, or 7 frames), suppressing high-frequency motion details and making the motion appear blurry and sluggish.

A.2 Randomized Segmented Corruption Application

To ensure comprehensive coverage of corruption scenarios, we adopt a segmented random application strategy. Each motion is divided into random temporal segments, and each segment is either kept clean or assigned one corruption type according to the predefined probability distribution.

Table 7 lists the complete parameter configuration for all corruption types used in our experiments. The scale factor in Table **??** is a normalized control coefficient rather than a unified physical quantity. Since different corruption types involve different underlying units, such as rotation, translation, duration, or contact displacement, scale values are only comparable within the same corruption mode. We will release the full implementation code to provide complete details of the corruption application process and parameterization.

B Details of MoreDiff Model

Input Format Our MoreDiff model is a standard Masked Motion Model enhanced with manifold rectification training to embed the step-agnostic Manifold Rectification Gradient (MRG) into the score function. Its inputs include the corrupted motion sequence (composed of 201-dimensional $\boldsymbol{\phi}_t$ vectors), diffusion step t, and an optional keyframe mask (same dimension as the motion input).

Model Architecture The MoreDiff model adopts a 12-layer Diffusion Transformer (DIT) with Rotary Position Embedding (RoPE), where $d_{\text{model}} = 768$, $n_{\text{head}} = 12$, and total parameters are 0.11B.

Frame Rate MoreDiff is configured to operate at 30 FPS. This setting is convenient for practical motion refinement because most video-based mocap sources are extracted from 30-FPS videos, while optical mocap data is typically captured at a frame rate that is an integer multiple of 30 FPS (e.g., 60, 120).

Inference Time Under the default inference settings ($T = 10$, use Ada-Rec), refining a 5-second motion sequence (150 frames at 30 FPS) on one GPU takes ≈ 0.2 seconds.

C Training Data Cleanup

To enable the model to effectively learn the Manifold Rectification Gradient (MRG) from corrupted motions to clean counterparts, we carefully clean the training data before corruption synthesis. Starting from approximately 38K motion sequences collected from AMASS, MOYO, LARa, FineDance, and Weizmann, we remove low-quality samples through a combination of physical-metric filtering and strict manual annotation. A sequence is marked as low quality if it contains visible joint pops, unnatural body twists, foot skating, or other motion artifacts that would make it unsuitable as clean supervision. In total, **3,533 low-quality sequences are excluded from the training set**. We then perform a second-stage review on these excluded samples and select 255 sequences with clear visual corruptions to construct the MoreDiff-test dataset for evaluating real optical mocap artifacts.

We will release our corruption annotations together with the code to facilitate future research on motion data cleanup and refinement.

D Rationale of Motion Representation Design

As defined in Sec. 3.1, MoreDiff represents each motion frame as

$$\boldsymbol{\phi}_n = \left[\boldsymbol{\theta}_n, \boldsymbol{p}_n, \boldsymbol{r}_n\right], \tag{19}$$

where $\boldsymbol{\theta}_n \in \mathbb{R}^{132}$ denotes the global orientations of 22 joints in the continuous 6D rotation format, $\boldsymbol{p}_n \in \mathbb{R}^{66}$ denotes root-relative 3D joint positions, and $\boldsymbol{r}_n \in \mathbb{R}^3$ denotes the global root translation. This representation is designed to preserve both local body geometry and global trajectory information during motion refinement.

The key design choice is to use **global joint orientations $\boldsymbol{\theta}_n$** instead of the commonly used local joint rotations. During diffusion q-sampling, perturbing local rotations can introduce **chained propagated orientation noise**: noise added to upstream joints is accumulated by downstream joints through the kinematic chain. As a result, distal joints may become over-noised even when the injected perturbation at each local joint is moderate, which degrades motion fidelity during refinement. By representing each joint directly with its global orientation, MoreDiff avoids this chained error propagation and provides better fidelity preservation.

We also avoid over-parameterized motion descriptors such as joint velocities and root angular velocities. Since $\boldsymbol{p}_n$ already encodes root-relative body geometry and $\boldsymbol{r}_n$ records the global trajectory, adding extra velocity terms increases feature dimensionality and corruption synthesis cost without being necessary for our refinement objective. The resulting representation remains compact while retaining the information required for both fidelity preservation and artifact correction.

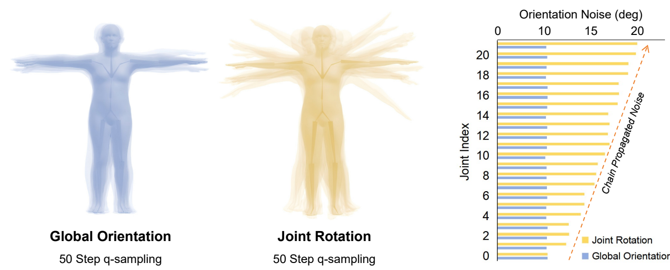

Fig. 11. Rationale for using global joint orientations in MoreDiff. Local joint rotations propagate noise along the kinematic chain during q-sampling, causing distal joints to accumulate larger orientation errors. Representing each joint directly by its global orientation avoids this chained noise amplification and improves refinement fidelity.

To validate the advantage of global joint orientations over local joint rotations, we conduct an ablation study by training an identical model architecture with local joint rotations replacing $\boldsymbol{\theta}_n$ while keeping $\boldsymbol{p}_n$ and $\boldsymbol{r}_n$ unchanged.

As shown in Table 9, the Global Orientation consistently outperforms Joint Rotation across all fidelity metrics under both denoise and ada-denoise modes. In particular, WA-MPJAE, which directly measures joint orientation accuracy, degrades significantly when replacing global joint orientations with local joint rotations, e.g., $3.59 \rightarrow 5.13$ under denoise-10 and $3.04 \rightarrow 4.23$ under ada-denoise-10. These results confirm that chained orientation noise can impair motion fidelity, and support our choice of using global orientation representation in MoreDiff.

E Variable-Length Support

To enable our frame-fixed model to handle arbitrary-length motion sequences, we employ a sliding-window inference pipeline with overlapping stitching to ensure temporal continuity. We use a 5-second (150-frame) window at 30fps for refinement, and slide it across the sequence with a small overlap (10 frames). To avoid discontinuities between adjacent windows, we perform three key operations: skeleton synchronization to eliminate bone-length drift, root orientation and position alignment to maintain global consistency, and linear blending in the overlapping region for smooth transition. We also propagate translation increments across windows to prevent global trajectory drifting. In the comparative experiments of this paper, this method is synchronously applied to the inference of StableMotion and CondMDI to ensure the fairness of the comparison. The overall pipeline is summarized in Alg. 2.

F Evaluation on BrokenAMASS

We further evaluate MoreDiff on BrokenAMASS, a synthetic corruption benchmark introduced by StableMotion [Mu et al. 2025]. BrokenAMASS is constructed from the same clean motion split as

Table 8. Quantitative Results on BrokenAMASS

Method	Fidelity Metrics (vs Clean)					Quality Metrics						
	MPJPE↓	WA-MPJPE↓	WA-MPJAE↓	RTE↓	Accel↓	FF↓	FP↓	FP Rate↓	FS↓	FS Rate↓	Jitter→	Pop Rate→
Clean	/	/	/	/	/	6.99	10.25	5.91	9.65	13.18	0.07	0.09
Corrupted Input	$1.63^{\pm 2.34}$	$4.22^{\pm 3.65}$	$3.81^{\pm 5.38}$	$3.16^{\pm 2.93}$	$0.89^{\pm 1.52}$	6.83	6.81	5.55	14.74	15.17	0.44	0.45
StableMotion	$3.88^{\pm 3.31}$	$17.67^{\pm 25.24}$	$11.36^{\pm 11.52}$	$16.39^{\pm 25.28}$	$1.53^{\pm 2.77}$	12.35	13.65	7.13	15.22	16.83	0.63	0.17
CondMDI-500	$3.16^{\pm 2.38}$	$12.32^{\pm 14.54}$	$8.15^{\pm 5.92}$	$11.55^{\pm 14.33}$	$0.30^{\pm 0.35}$	6.88	10.10	8.18	9.51	13.75	0.06	0.03
CondMDI-250	$2.52^{\pm 2.09}$	$8.75^{\pm 9.81}$	$6.47^{\pm 5.29}$	$8.09^{\pm 9.44}$	$0.28^{\pm 0.34}$	6.94	8.60	7.28	9.82	14.25	0.06	0.03
CondMDI-100	$2.11^{\pm 2.02}$	$6.53^{\pm 7.05}$	$5.38^{\pm 5.10}$	$5.90^{\pm 6.52}$	$0.29^{\pm 0.34}$	7.11	7.67	6.59	10.39	15.05	0.06	0.03
CondMDI-10	$1.78^{\pm 2.15}$	$4.98^{\pm 4.98}$	$4.43^{\pm 5.16}$	$4.25^{\pm 4.29}$	$0.38^{\pm 0.49}$	7.65	6.99	5.01	11.90	17.19	0.10	0.04
MoreDiff (T=10)	$1.55^{\pm 2.13}$	$7.56^{\pm 13.03}$	$3.86^{\pm 5.01}$	$7.04^{\pm 12.65}$	$0.31^{\pm 0.40}$	7.68	8.08	4.07	11.00	16.42	0.07	0.03

Table 9. Ablation Study on Motion Representation: Global Joint Orientation (Full Model) vs. Local Joint Rotation (A2). All models are evaluated on CorruptedAMASS with 10-step diffusion rectification. ↓: lower is better.

Representation	**Mode**	MPJPE↓	WA-MPJPE↓	WA-MPJAE↓	RTE↓	Accel↓
Joint Rotation	denoise-10	$1.67^{\pm 1.15}$	$7.30^{\pm 10.54}$	$5.13^{\pm 3.71}$	$6.82^{\pm 10.52}$	$0.23^{\pm 0.22}$
Global Orientation	denoise-10	$1.27^{\pm 1.27}$	$5.78^{\pm 11.04}$	$3.59^{\pm 4.38}$	$5.45^{\pm 10.93}$	$0.21^{\pm 0.21}$
Joint Rotation	ada-denoise-10	$1.41^{\pm 1.31}$	$7.19^{\pm 12.66}$	$4.23^{\pm 4.30}$	$6.78^{\pm 12.57}$	$0.21^{\pm 0.22}$
Joint Orientation, 0.1B	ada-denoise-10	$\mathbf{1.05}^{\pm 1.31}$	$\mathbf{5.48}^{\pm 13.43}$	$\mathbf{3.04}^{\pm 4.50}$	$\mathbf{5.19}^{\pm 13.29}$	$\mathbf{0.19}^{\pm 0.20}$

ALGORITHM 2: Sliding-Window Refinement for Variable-Length Motion

Input: Full-length motion $\mathcal{M}$, window size $N = 150$, overlap $O = 10$
Output: Refined full-length motion $\mathcal{M}^*$
$current \leftarrow 0$, $prev\_pad \leftarrow 0$
while $current < \mathrm{len}(\mathcal{M})$ **do**
  $end \leftarrow \min(current + N, \mathrm{len}(\mathcal{M}))$
  $\mathcal{W} \leftarrow \mathcal{M}[:, current : end]$
  $\mathrm{Mask}[:, 0 : prev\_pad] \leftarrow 1$ {Keep overlap region unchanged}
  $\mathcal{W}^* \leftarrow \mathrm{MoreDiffRefine}(\mathcal{W}, \mathrm{Mask})$ {Refine non-overlap frames}
  $\mathcal{W}^* \leftarrow \mathrm{SyncSkeleton}(\mathcal{W}^*)$ {Fix bone length consistency}
  $\mathcal{W}^* \leftarrow \mathrm{AlignRoot}(\mathcal{W}^*)$ {Align orientation & position}
  if $prev\_pad > 0$ **then**
    $\mathcal{W}^* \leftarrow \mathrm{LinearBlend}(\mathcal{W}^*, prev\_pad)$ {Smooth overlapping frames}
  end if
  $\Delta t \leftarrow \mathcal{W}^*[-1] - \mathcal{M}[end - 1]$ {Compute translation delta}
  $\mathcal{M}[:, current : end] \leftarrow \mathcal{W}^*$
  $\mathcal{M}[:, end :] \leftarrow \mathcal{M}[:, end :] + \Delta t$ {Propagate global translation}
  $current \leftarrow current + N - O$
  $prev\_pad \leftarrow O$
end while
return $\mathcal{M}^* = \mathcal{M}$

CorruptedAMASS, but uses a different corruption synthesis pipeline and therefore contains different corruption patterns. This setting allows us to test whether MoreDiff overfits to our own corruption synthesis strategy, or instead learns a more general manifold rectification behavior that can transfer to shifted synthetic corruption distributions. Since paired clean references are available, fidelity metrics are computed against the clean motions.

As shown in Table 8, although MoreDiff achieves the best MPJPE and restores Jitter to the clean-motion level, it does not obtain the best WA-MPJPE or RTE on BrokenAMASS. This indicates that unseen corruption patterns, particularly root-level and trajectory-related artifacts from a different synthesis pipeline, can still reduce refinement performance. The result highlights the importance of enriching synthetic corruption patterns for improving the robustness of manifold rectification. In future work, we will analyze corruption characteristics from more motion capture systems, including inertial motion capture and ego-camera-based capture, and incorporate them into our synthesis pipeline to further broaden the application scope of MoreDiff.